Unit 10, Lesson 6: Comparing Data Sets

Benchmark

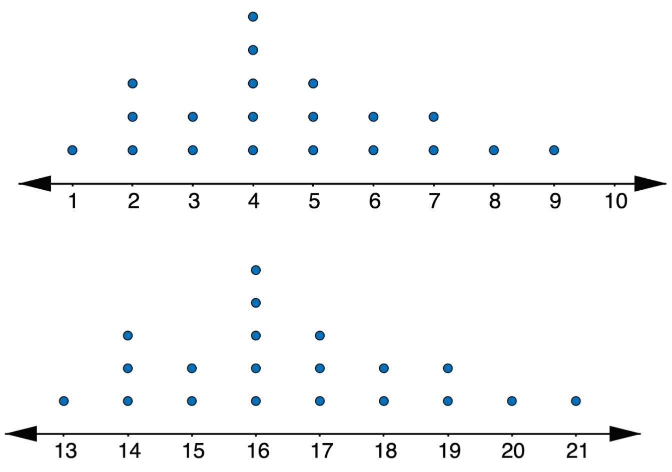
MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

Learning Target

☐ I can compare and interpret two numerical representations of data for two populations.

10.6.1 Warm-Up: Notice and Wonder

The line plots represent the distribution of the tip amounts, in dollars, left at two different restaurants on the same night.



Tip Amount (in dollars)

1. Complete each statement.

   a. When looking at the data displays, one thing I notice is …

   b. One thing I wonder is …

10.6.2 Exploration: Calculating Measures of Center and Variation

On a college entrance exam, the possible scores on the math section range from 200-800. Below are the scores for 20 randomly selected female high school seniors.

454 409 437 415 452 440 454 478 401 403 426 426 437 465 465 429 470 448 511 453

The scores for 20 randomly selected male high school seniors are shown below.

408 440 414 391 480 415 426 392 469 446 409 432 375 388 413 451 370 415 487 404

1. Work with your partner to complete the table by calculating the measures of center and variability for each set of data.

Measure	Female Students	Male Students
Mean		
Median		
Range		
IQR		
Potential Outliers		

2. Complete the statements.

   a. Female students tend to have ○ lower ○ higher scores on the math section of the college entrance exam, with a median of ________, than males, who have a median of ________.

   b. Scores for males tend to have ○ less ○ more variability, with an IQR of ________, than scores for females, with an IQR of ________.

   c. Explain why the measures median and IQR were chosen to describe the data sets above instead of mean and range.

   d. Check your work on Parts A-C with your partner and, if necessary, resolve any differences.

10.6.3 Try It: Drawing Conclusions

Shawn wants to determine the best way to get to school. Over several weeks, he took turns riding in a car and riding his bicycle and recorded the time, in minutes.

Car	
19	17
15	18
12	22
29	14

Bicycle	
24	25
27	26
21	28
32	23

Shawn calculated the measures of center and spread for each transportation method. They are shown in the table.

	Car	Bicycle
Mean	18.25	25.75
Median	17.5	25.5
Range	17.0	11.0
IQR	6.0	4.0

1. Make an argument as to why Shawn should ride in a car to get to school.

2. Make an argument as to why Shawn should ride his bicycle to school.

10.6.4 Activity: Guess My Data Sets

Suppose an English teacher takes five random samples of students and records their most recent test scores.

Data Set A: $\{22, 0, 44, 40, 56, 32, 64, 44, 52, 60\}$

Data Set B: $\{60, 62, 62, 60, 64, 68, 64, 62, 68, 60\}$

Data Set C: $\{70, 82, 74, 76, 80, 72, 84, 70, 78, 82\}$

Data Set D: $\{72, 86, 100, 76, 92, 90, 100, 78, 98, 86\}$

Data Set E: $\{64, 36, 74, 82, 96, 80, 66, 78, 68, 0\}$

Without telling your partner, choose two data sets from above.

1. Write a paragraph comparing the two data sets you have chosen using measures of center and variability. Make sure to refer to the data sets as “Data Set 1” and “Data Set 2” in the paragraph you write.

2. Determine with your partner who is partner A and who is partner B. Write your names on the lines for each partner in the table.

3. Partner A should read their paragraph from Number 1 to partner B, and partner B will guess which two data sets were used. Then, partner B will explain their reasoning for their guess. Both partners will summarize the guess and reasoning in the first row of the table.

4. Then, switch roles so that partner B reads their paragraph while partner A makes a guess and explains their reasoning.

Partner	Guess	Reasoning for Guess
Partner A: ____________		
Partner B: ____________		

5. Reveal to your partner which data sets were used and discuss any errors that may have led to incorrect conclusions.

10.6.5 Your Turn!

1. A company is trying to determine which method of communication is best to tell their employees important information. They send text messages to nine employees and emails to another nine employees. They record how many minutes it takes for the employees to reply.

Text	5	20	45	9	13	12	4	1	11
Email	25	15	35	24	30	19	29	38	12

a. Which measure of center best summarizes the data sets?

Text: ○ mean ○ median

Email: ○ mean ○ median

b. Which measure of variability best summarizes the data sets?

Text: ○ range ○ IQR

Email: ○ range ○ IQR

c. Calculate the measures of center and variability.

	Text	Email
Mean		
Median		
Range		
IQR		

d. Use the most appropriate measures of center and variability to compare the effectiveness of the two methods of communicating with employees.

10.6.6 Wrap-Up: Cool Down

1. Use the data from Your Turn to complete the prompt below.

   Lucas recommends that the company emails its employees because of the smaller range in the data set. Explain why you agree or disagree with Lucas.